

Color Segmentation

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Preface

This tutorial applies to firmware version: CanMV_K230_YAHBOOM_micropython_v1.8-13. You can download it from the Jianan website https://download.kendryte.com/developer/releases/canmv_k230_micropython/daily_build/

 **K230 MicroPython 镜像下载**

CI/CD 更新时间: 2026-07-28 01:29:59

选择开发板

 CanMV K230 V1.0/V1.1

 CanMV K230 V3.0

 庐山派 K230

 庐山派 Lite K230D

 **Yahboom K230**

 01Studio K230

 01Studio K230 Emmc

 Hiwonder K230

 WonderMK K230

 GT6700 K230

 **Yahboom K230**
MicroPython Build for Yahboom K230

当前下载镜像:
[CanMV_K230_YAHBOOM_micropython_v1.8-28-g9b151af_nncase_v2.11.0.img.gz](#)

最新每日构建版本
Daily-0728
编译日期: 2026-07-28

MD5: 66f0622fc6b8d22e55f552a40458d2fa
立即下载

极简 USB 烧录指南 (推荐)

⚠ 镜像仅可烧录到对应的开发板，请务必选择与当前板型完全匹配的镜像，不能混烧其他板型的镜像。否则可能导致硬件损坏!!!

1 准备工具
下载并打开 [K230BurningTool](#) 烧录工具，并加载固件。

2 进入烧录模式 (根据硬件情况选择)

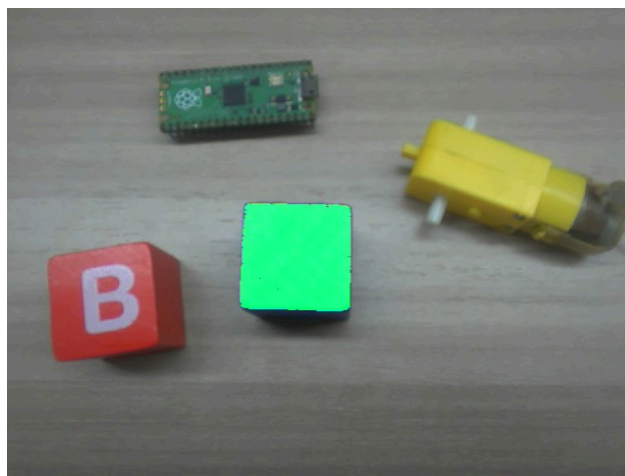
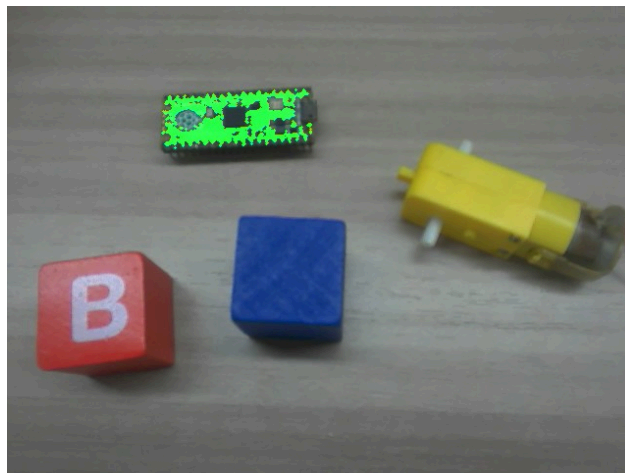
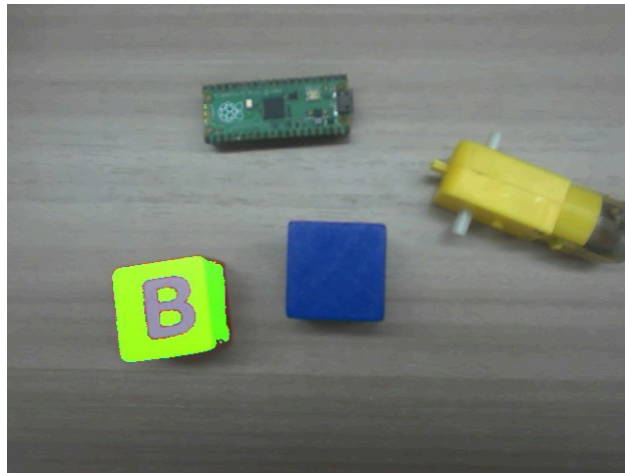
☒ 有 BOOT 键的板子

☐ 无 BOOT 键的板子

Example Results

Run the example code for this section [13.camera inRange color.py](#).

This section converts live BGR images to HSV space, uses `inRange()` to segment red, green, blue, and yellow colors, applies opening operation to remove small noise, and overlays the mask onto the green channel to create a highlight effect.





Principle Explanation

HSV Threshold Segmentation

HSV separates hue, saturation, and value. `inRange()` checks whether each pixel's three components are simultaneously between the lower and upper limits, outputting 255 within the range and 0 outside.

The thresholds used in the source code are as follows:

<code>\$mode\$</code>	Color	H Range	S Range	V Range
0	Red	0~10	100~255	100~255
1	Green	35~85	60~255	60~255
2	Blue	100~130	100~255	100~255
3	Yellow	20~35	100~255	100~255

The S and V lower limits can exclude low-saturation grayish-white areas and overly dark pixels.

Red Hue Wraparound

In 8-bit OpenCV HSV, H is typically 0~179. Red is at both ends of the hue axis. The source code only covers the low hue segment of 0~10, so reddish-purple or red in the 170~179 range may be missed. Complete red segmentation should establish two mask segments and merge them.

Code Overview

Initialize Camera and Display Module

```
w, h = 640, 480
sensor = Sensor(width=1280, height=960, fps=90)
sensor.reset()
sensor.set_framesize(width=w, height=h)
sensor.set_pixformat(Sensor.RGB888)
Display.init(Display.ST7701, width=640, height=480, to_ide=True)
sensor.run()
```

```
time.sleep(0.5)
```

Set Color Thresholds and Mode

```
cth = [  
    [0, 100, 100, 10, 255, 255],  
    [35, 60, 60, 85, 255, 255],  
    [100, 100, 100, 130, 255, 255],  
    [20, 100, 100, 35, 255, 255]  
]  
mode = 0  
k = cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (3, 3))
```

The source code defaults to detecting red. The morphological kernel is created outside the main loop to avoid per-frame reallocation.

Convert Color and Generate Mask

```
hsv = cv2.cvtColor(img_np, cv2.COLOR_BGR2HSV)  
threshold = cth[mode]  
low = np.array([threshold[0], threshold[1], threshold[2]])  
high = np.array([threshold[3], threshold[4], threshold[5]])  
mask = cv2.inRange(hsv, low, high)
```

Both upper and lower limit arrays contain H, S, and V components. If any component is outside the range, the output mask pixel is 0.

Opening Operation for Denoising

```
mask = cv2.morphologyEx(mask, cv2.MORPH_OPEN, k)
```

The 3×3 elliptical kernel removes small white spots. If the target region has internal holes, you can add closing operation after opening, but it increases computation.

Highlight Target Region

```
img_np[:, :, 1] = cv2.add(img_np[:, :, 1], mask)  
Display.show_image(img)
```

The mask is added to channel 1, making detected regions more greenish. `cv2.add()` uses saturation addition; results exceeding 255 remain 255 without 8-bit wraparound.

Resource Release

```
finally:  
    sensor.stop()  
    Display.deinit()  
    os.exitpoint(os.EXITPOINT_ENABLE_SLEEP)  
    time.sleep_ms(100)
```

Parameter Adjustment Notes

- **Sample First Then Set Threshold:** Read the target HSV under actual camera, lighting, and material conditions, then set the range around the sample.
- **S Lower Limit:** Increasing can exclude grayish-white background but may miss low-saturation targets.
- **V Lower Limit:** Increasing can exclude shadows and dark areas but leads to more missed detections in low light.
- **Red Dual Range:** Call `inRange()` separately for 0~10 and 170~179, then merge with `bitwise_or()`.
- **Morphological Kernel:** When noise points are larger than 3×3, you can use 5×5, but confirm it does not erode small targets.
- **Result Statistics:** When you need area or target count, use `countNonZero()`, `findContours()`, or connected component analysis on mask.